

EXPLORATORY DATA ANALYSIS (EDA)

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: df = pd.read_csv(r"world_population.csv")
df
```

Out[2]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	Pc
0	36	AFG	Afghanistan	Kabul	Asia	41128771.0	38972230.0	33753499.0	28
1	138	ALB	Albania	Tirana	Europe	2842321.0	2866849.0	2882481.0	2
2	34	DZA	Algeria	Algiers	Africa	44903225.0	43451666.0	39543154.0	35
3	213	ASM	American Samoa	Pago Pago	Oceania	44273.0	46189.0	51368.0	
4	203	AND	Andorra	Andorra la Vella	Europe	79824.0	77700.0	71746.0	
...	...	...	...	...	...	...	...	...	...
229	226	WLF	Wallis and Futuna	Mata-Utu	Oceania	11572.0	11655.0	12182.0	
230	172	ESH	Western Sahara	El Aaiún	Africa	575986.0	556048.0	491824.0	
231	46	YEM	Yemen	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24
232	63	ZMB	Zambia	Lusaka	Africa	20017675.0	18927715.0	NaN	13
233	74	ZWE	Zimbabwe	Harare	Africa	16320537.0	15669666.0	14154937.0	12

234 rows × 17 columns



```
In [5]: pd.set_option('display.float_format', lambda x: '%.2f' % x) #upto 2 decimal
```

```
In [6]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 234 entries, 0 to 233
Data columns (total 17 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Rank                                  234 non-null    int64
 1   CCA3                                  234 non-null    str
 2   Country                              234 non-null    str
 3   Capital                              234 non-null    str
 4   Continent                            234 non-null    str
 5   2022 Population                      230 non-null    float64
 6   2020 Population                      233 non-null    float64
 7   2015 Population                      230 non-null    float64
 8   2010 Population                      227 non-null    float64
 9   2000 Population                      227 non-null    float64
10   1990 Population                      229 non-null    float64
11   1980 Population                      229 non-null    float64
12   1970 Population                      230 non-null    float64
13   Area (km²)                          232 non-null    float64
14   Density (per km²)                   230 non-null    float64
15   Growth Rate                         232 non-null    float64
16   World Population Percentage          234 non-null    float64
dtypes: float64(12), int64(1), str(4)
memory usage: 37.5 KB

```

In [7]: `df.describe()` *# will give values like mean, min, max, count, std*

Out[7]:

	Rank	2022 Population	2020 Population	2015 Population	2010 Population	2000 Population
count	234.00	230.00	233.00	230.00	227.00	227.00
mean	117.50	34632250.88	33600710.95	32066004.16	30270164.48	26840495.26
std	67.69	137889172.44	135873196.61	131507146.34	126074183.54	113352454.57
min	1.00	510.00	520.00	564.00	596.00	651.00
25%	59.25	419738.50	406471.00	394295.00	382726.50	329470.00
50%	117.50	5762857.00	5456681.00	5244415.00	4889741.00	4491202.00
75%	175.75	22653719.00	21522626.00	19730853.75	16825852.50	15625467.00
max	234.00	1425887337.00	1424929781.00	1393715448.00	1348191368.00	1264099069.00

In [8]: `df.isnull().sum()` *# total null or empty cell in each columns*

```
Out[8]: Rank          0
        CCA3          0
        Country       0
        Capital        0
        Continent      0
        2022 Population 4
        2020 Population 1
        2015 Population 4
        2010 Population 7
        2000 Population 7
        1990 Population 5
        1980 Population 5
        1970 Population 4
        Area (km²)      2
        Density (per km²) 4
        Growth Rate     2
        World Population Percentage 0
        dtype: int64
```

```
In [10]: df.nunique()    # unique values in each columns
```

```
Out[10]: Rank          234
        CCA3          234
        Country       234
        Capital        234
        Continent        6
        2022 Population 230
        2020 Population 233
        2015 Population 230
        2010 Population 227
        2000 Population 227
        1990 Population 229
        1980 Population 229
        1970 Population 230
        Area (km²)      231
        Density (per km²) 230
        Growth Rate     178
        World Population Percentage 70
        dtype: int64
```

```
In [11]: df.sort_values(by="World Population Percentage", ascending=False).head(10)
```

Out[11]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	P
41	1	CHN	China	Beijing	Asia	1425887337.00	1424929781.00	1393
92	2	IND	India	New Delhi	Asia	1417173173.00	1396387127.00	1322
221	3	USA	United States	Washington, D.C.	North America	338289857.00	335942003.00	324
93	4	IDN	Indonesia	Jakarta	Asia	275501339.00	271857970.00	259
156	5	PAK	Pakistan	Islamabad	Asia	235824862.00	227196741.00	210
149	6	NGA	Nigeria	Abuja	Africa	218541212.00	208327405.00	183
27	7	BRA	Brazil	Brasilia	South America	215313498.00	213196304.00	205
16	8	BGD	Bangladesh	Dhaka	Asia	171186372.00	167420951.00	157
171	9	RUS	Russia	Moscow	Europe	144713314.00	145617329.00	144
131	10	MEX	Mexico	Mexico City	North America	127504125.00	125998302.00	120

In [13]: `df.corr(numeric_only=True)`

Out[13]:

	Rank	2022 Population	2020 Population	2015 Population	2010 Population	2000 Population	1990 Population
Rank	1.00	-0.36	-0.36	-0.35	-0.35	-0.34	-0.33
2022 Population	-0.36	1.00	1.00	1.00	1.00	0.99	0.99
2020 Population	-0.36	1.00	1.00	1.00	1.00	1.00	0.99
2015 Population	-0.35	1.00	1.00	1.00	1.00	1.00	0.99
2010 Population	-0.35	1.00	1.00	1.00	1.00	1.00	1.00
2000 Population	-0.34	0.99	1.00	1.00	1.00	1.00	1.00
1990 Population	-0.33	0.99	0.99	0.99	1.00	1.00	1.00
1980 Population	-0.33	0.99	0.99	0.99	0.99	1.00	1.00
1970 Population	-0.34	0.97	0.98	0.98	0.98	0.99	1.00
Area (km²)	-0.38	0.45	0.45	0.46	0.46	0.47	0.52
Density (per km²)	0.13	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03
Growth Rate	-0.22	-0.02	-0.03	-0.03	-0.04	-0.05	-0.07
World Population Percentage	-0.36	1.00	1.00	1.00	1.00	0.99	0.99



```
In [15]: sns.heatmap(df.corr(numeric_only=True), annot = True)

plt.rcParams['figure.figsize'] = (20,7)

plt.show()
```



```
In [26]: df.groupby('Continent').mean(numeric_only=True).sort_values(by="2022 Population",as
# grouping, mean, sorting
```

Out[26]:

	Rank	2022 Population	2020 Population	2015 Population	2010 Population	2000 Population	Popu
Continent							
Asia	77.56	96327387.31	94955134.37	89165003.64	89087770.00	80580835.11	486395
South America	97.57	31201186.29	30823574.50	29509599.71	26789395.54	25015888.69	212247
Africa	92.16	25455879.68	23871435.26	21419703.57	18898197.31	14598365.95	113765
Europe	124.50	15055371.82	14915843.92	15027454.12	14712278.68	14817685.71	147852
North America	160.93	15007403.40	14855914.82	14259596.25	13568016.28	12151739.60	105316
Oceania	188.52	2046386.32	1910148.96	1756664.48	1613163.65	1357512.09	11627

```
In [27]: df[df['Continent'].str.contains('Oceania')] # filtering out
```

Out[27]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2011 Population
3	213	ASM	American Samoa	Pago Pago	Oceania	44273.00	46189.00	51368.00
11	55	AUS	Australia	Canberra	Oceania	26177413.00	25670051.00	23820236.00
44	223	COK	Cook Islands	Avarua	Oceania	17011.00	17029.00	17695.00
66	162	FJI	Fiji	Suva	Oceania	929766.00	920422.00	917200.00
70	183	PYF	French Polynesia	Papeete	Oceania	306279.00	301920.00	291787.00
81	191	GUM	Guam	Hagåtña	Oceania	171774.00	169231.00	167978.00
107	192	KIR	Kiribati	Tarawa	Oceania	131232.00	126463.00	116707.00
126	215	MHL	Marshall Islands	Majuro	Oceania	41569.00	43413.00	49410.00
132	194	FSM	Micronesia	Palikir	Oceania	114164.00	112106.00	109462.00
142	225	NRU	Nauru	Yaren	Oceania	12668.00	12315.00	11185.00
145	185	NCL	New Caledonia	Nouméa	Oceania	289950.00	286403.00	283032.00
146	123	NZL	New Zealand	Wellington	Oceania	5185288.00	5061133.00	4590590.00
150	232	NIU	Niue	Alofi	Oceania	1934.00	1942.00	1847.00
153	210	NFK	Northern Mariana Islands	Saipan	Oceania	49551.00	49587.00	51514.00
157	222	PLW	Palau	Ngerulmud	Oceania	NaN	17972.00	17794.00
160	93	PNG	Papua New Guinea	Port Moresby	Oceania	10142619.00	9749640.00	8682174.00
179	188	WSM	Samoa	Apia	Oceania	222382.00	214929.00	203571.00
191	166	SLB	Solomon Islands	Honiara	Oceania	724273.00	691191.00	612660.00
209	233	TKL	Tokelau	Nukunonu	Oceania	1871.00	1827.00	1454.00
210	197	TON	Tonga	Nuku'alofa	Oceania	106858.00	105254.00	106122.00
216	227	TUV	Tuvalu	Funafuti	Oceania	11312.00	11069.00	10877.00
225	181	VUT	Vanuatu	Port-Vila	Oceania	326740.00	311685.00	276438.00
229	226	WLF	Wallis and Futuna	Mata-Utu	Oceania	11572.00	11655.00	12182.00

```
In [19]: df2 = df.groupby('Continent')[['1970 Population',
    '1980 Population', '1990 Population', '2000 Population',
    '2010 Population', '2015 Population', '2020 Population',
    '2022 Population']].mean().sort_values(by="2022 Population", ascending=False)
df2
```

```
Out[19]:
```

	1970 Population	1980 Population	1990 Population	2000 Population	2010 Population	2015 Population
Continent						
Asia	43839877.83	40278333.33	48639995.33	80580835.11	89087770.00	89165003.64
South America	13781939.71	17270643.29	21224743.93	25015888.69	26789395.54	29509599.71
Africa	6567175.27	8586031.98	11376964.52	14598365.95	18898197.31	21419703.57
Europe	13118479.82	14200004.52	14785203.94	14817685.71	14712278.68	15027454.12
North America	7885865.15	9207334.03	10531660.62	12151739.60	13568016.28	14259596.25
Oceania	846968.26	996532.17	1162774.87	1357512.09	1613163.65	1756664.48

```
In [20]: df.columns
```

```
Out[20]: Index(['Rank', 'CCA3', 'Country', 'Capital', 'Continent', '2022 Population',
    '2020 Population', '2015 Population', '2010 Population',
    '2000 Population', '1990 Population', '1980 Population',
    '1970 Population', 'Area (km²)', 'Density (per km²)', 'Growth Rate',
    'World Population Percentage'],
    dtype='str')
```

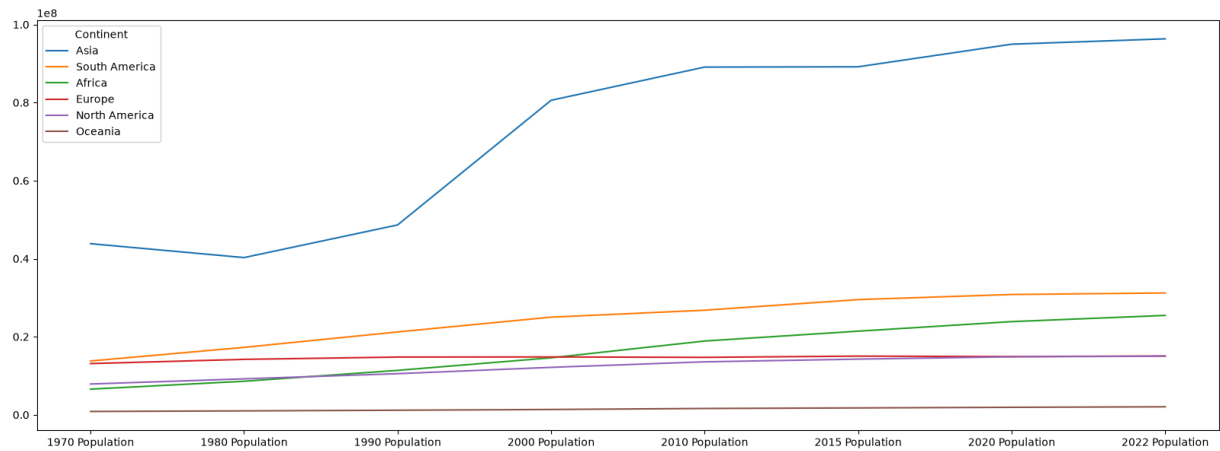
```
In [21]: df3 = df2.transpose()
df3
```

Out[21]:

Continent	Asia	South America	Africa	Europe	North America	Oceania
1970 Population	43839877.83	13781939.71	6567175.27	13118479.82	7885865.15	846968.26
1980 Population	40278333.33	17270643.29	8586031.98	14200004.52	9207334.03	996532.17
1990 Population	48639995.33	21224743.93	11376964.52	14785203.94	10531660.62	1162774.87
2000 Population	80580835.11	25015888.69	14598365.95	14817685.71	12151739.60	1357512.09
2010 Population	89087770.00	26789395.54	18898197.31	14712278.68	13568016.28	1613163.65
2015 Population	89165003.64	29509599.71	21419703.57	15027454.12	14259596.25	1756664.48
2020 Population	94955134.37	30823574.50	23871435.26	14915843.92	14855914.82	1910148.96
2022 Population	96327387.31	31201186.29	25455879.68	15055371.82	15007403.40	2046386.32

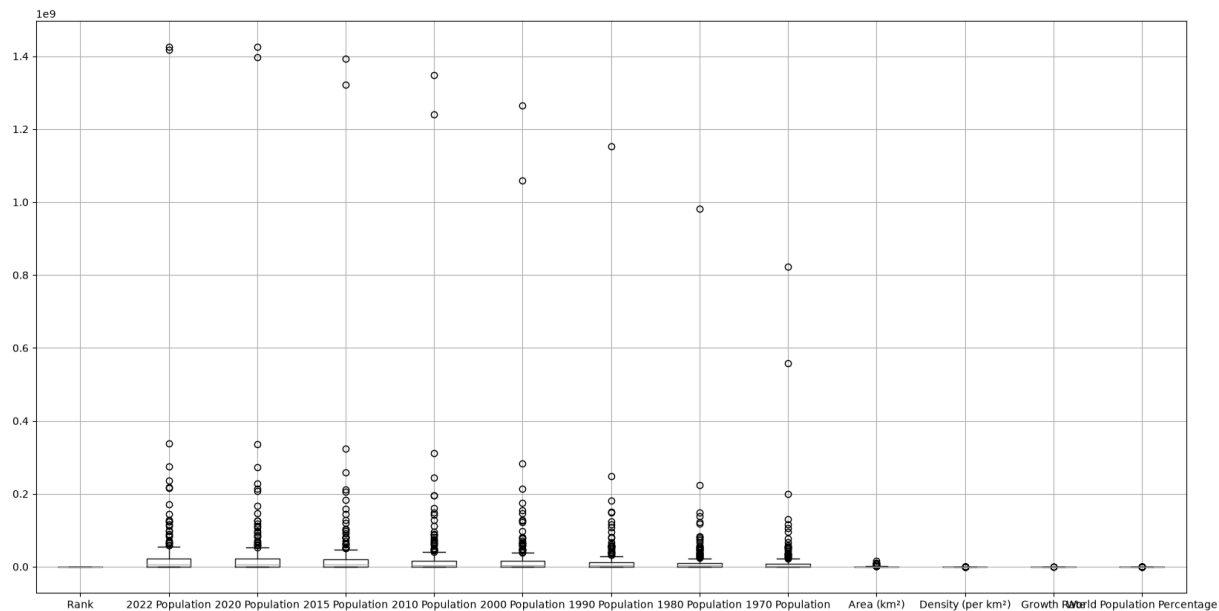
In [22]: df3.plot()

Out[22]: <Axes: >



In [23]: df.boxplot(figsize=(20,10))

Out[23]: <Axes: >



```
In [25]: df.select_dtypes(include='float') # including only floats
```

Out[25]:

	2022 Population	2020 Population	2015 Population	2010 Population	2000 Population	1990 Population	Popul
0	41128771.00	38972230.00	33753499.00	28189672.00	19542982.00	10694796.00	124866
1	2842321.00	2866849.00	2882481.00	2913399.00	3182021.00	3295066.00	29416
2	44903225.00	43451666.00	39543154.00	35856344.00	30774621.00	25518074.00	187393
3	44273.00	46189.00	51368.00	54849.00	58230.00	47818.00	328
4	79824.00	77700.00	71746.00	71519.00	66097.00	53569.00	356
...	...	...	...	...	...	...	...
229	11572.00	11655.00	12182.00	13142.00	14723.00	13454.00	113
230	575986.00	556048.00	491824.00	413296.00	270375.00	178529.00	1167
231	33696614.00	32284046.00	28516545.00	24743946.00	18628700.00	13375121.00	92049
232	20017675.00	18927715.00	NaN	13792086.00	9891136.00	7686401.00	57204
233	16320537.00	15669666.00	14154937.00	12839771.00	11834676.00	10113893.00	70499

234 rows × 12 columns

